Song Xue

Ph.D. Candidate in Pathogen Biology · Fudan University

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Research Interests

Pathogen biology and humoral immunity, focusing on antibody discovery, screening, and functional characterization. Research integrates high-throughput single B cell screening, deep sequencing, and single-cell multi-omics to study how antibody sequence and structure influence broad neutralizing activity and antibody-dependent enhancement (ADE).

Education

Sep 2022 – Present **Ph.D. in Pathogen Biology**, *Fudan University*

Sep 2017 – Jun 2022 **B.Sc. in Forensic Medicine**, *Fudan University*

Research Experience

High-Throughput Screening and Functional Validation of Broad-Spectrum Anti-SARS-CoV-2 Neutralizing Antibodies

Developed and optimized a high-throughput affinity evaluation platform for multiplexed antigen–antibody screening based on 10x single-cell transcriptome–BCR–surface antigen co-sequencing (LIBRA-seq).

Identified multiple monoclonal antibodies with ultra-broad binding to the receptor-binding domains of diverse coronaviruses.

Epitope-Specific Mechanisms of Antibody-Dependent Enhancement (ADE) Across SARS-CoV-2 Variants

Used structurally characterized monoclonal antibodies to investigate how epitope specificity drives distinct ADE phenotypes.

Integrated bulk RNA-seq and scRNA-seq data to compare ADE effects and host immune responses across SARS-CoV-2 variants.

Compared the ADE effects of antibodies from COVID-19 vaccine recipients and convalescent individuals.

Isolation and Identification of Fully Human Monoclonal Antibodies Targeting HBV Core/e Antigens

Established and optimized a prokaryotic expression system for the in vitro recombinant production of HBV core and e antigens.

Isolated multiple fully human monoclonal antibodies from HBV-convalescent individuals that efficiently recognize both antigens.

High-Resolution Profiling of HBV Quasispecies and Immune Escape in Chronic Hepatitis B

Applied MutPE-seq, a high-fidelity paired-end deep sequencing strategy, to improve the detection resolution of low-frequency HBV variants to 0.5%.

Profiled 35 patients with chronic hepatitis B at high resolution, linking low-frequency mutations and genomic heterogeneity to clinical phenotype, viral replication fitness, and antibody therapy response.

Publications

Xue S., Han Y., Wu F., Wang Q. Mutations in the SARS-CoV-2 spike receptor binding domain and their delicate balance between ACE2 affinity and antibody evasion. *Protein & Cell*. 2024. (First author)

Han Y., Xue S., Xu W., Zhang Y., Peng X., et al. Divergent antibody-enhanced entry across SARS-CoV-2 lineages drives variant-specific immunopathology through epistatically constrained spike evolution. *LangTaoSha Preprint Server*. 2025. Preprint. (Co-first author, second author)

Peng X., Han Y., Xue S., Zhou Y., Jiang W., et al. Low Antibody-Dependent Enhancement of Viral Entry Activity Supports the Safety of Inactivated SARS-CoV-2 Vaccines. *Vaccines*. 2025. (Co-first author)

Wu J., Chen Z., Gao Y., Wang Z., Wang J., et al. Fortuitous somatic mutations during antibody evolution endow broad neutralization against SARS-CoV-2 Omicron variants. *Cell Reports*. 2023. (Co-author)

Xia A., Wang X., He J., Wu W., Jiang W., Xue S., et al. Cross-reactive antibody response to Monkeypox virus surface proteins in a small proportion of individuals with and without Chinese smallpox vaccination history. *BMC Biology*. 2023. (Co-author)

Sun S., He J., Liu L., Zhu Y., Zhang Q., et al. Anti-S2 antibodies responsible for the SARS-CoV-2 infection-induced serological cross-reactivity against MERS-CoV and MERS-related coronaviruses. *Frontiers in Immunology*. 2025. (Co-author)

Jiang W., Wu J., He J., Xia A., Wu W., et al. Re-Evaluation and Retrospective Comparison of Serum Neutralization Induced by Three Different Types of Inactivated SARS-CoV-2 Vaccines. *Vaccines*. 2024. (Co-author)

Honors & Awards

Outstanding Doctoral Student Academic Scholarship, Fudan University (2026)
Outstanding Student, Fudan University
Eisai (China) Scholarship
One Health Foundation Scholarship

Technical Skills

Antibody Discovery & Functional Characterization: Single B cell flow sorting; human/mouse PBMC isolation and magnetic bead separation; library design for single-cell transcriptome, BCR, and surface protein co-sequencing; nested PCR antibody sequence amplification; antibody production; competitive ELISA; neutralization assays; ADE functional testing; animal model construction and in vivo therapeutic evaluation; full workflow from antibody screening through expression, characterization, and validation.
Protein, Molecular & Cell Biology: Protein expression and purification (eukaryotic/prokaryotic recombinant expression; ÄKTA and gravity-column affinity chromatography); antigen and antibody characterization by ELISA, BLI, and AUC.
Bioinformatics & Single-Cell Analysis: QC, alignment, quantification, and variant calling for bulk RNA-seq and targeted deep sequencing; single-cell clustering and dimensionality reduction; marker and differential gene identification; cell proportion and phenotype analysis; viral sequence alignment and variant analysis.

Academic Activities

International exchange, University of Duisburg-Essen, Germany (online 2022; on-site 2025), with academic presentations.
Keynote speaker, Genewiz Single-Cell and Multi-Omics Innovation Day (July 2026), on high-throughput sequencing for antibody screening.